Tournaments not inducible by five voters

Leonid Chindelevitch[1] **and Ararat Harutyunyan**[2]

[1] MRC Centre for Global Infectious Disease Analysis, School of Public Health, Imperial College London, London, United Kingdom. Email: `lchindel@ic.ac.uk`. ORCID: 0000-0002-6619-6013.

[2] University of Paris-Dauphine, PSL University, CNRS UMR7243, LAMSADE, Paris, France. Email: `ararat.harutyunyan@lamsade.dauphine.fr`.

Corresponding author: Leonid Chindelevitch (`lchindel@ic.ac.uk`).

Abstract

How many voters must a committee have before its majority preferences can take any form at all? A tournament is k-inducible if some k linear orders produce it as their pairwise-majority relation, and for odd k we write $N(k)$ for the least number of alternatives at which some tournament is not k-inducible. Only $N(3) = 8$ is known exactly. For five voters the best bounds were $12 \leq N(5) \leq 38$, and the smallest explicit tournament beyond five voters had 43 vertices.

We improve both bounds to $13 \leq N(5) \leq 22$. The upper bound comes from two tournaments on 22 vertices, each a doubly regular tournament on 23 vertices with one vertex deleted, that five voters cannot produce; one of the two refutations is formally certified. On the way we show that the Paley tournament on 23 vertices is not 5-inducible, a case a general-purpose SAT solver had left open after six weeks, and that no doubly regular tournament on 23 vertices is. The lower bound comes from an analysis of all tournaments on 12 vertices. We also show that the Paley tournament on 19 vertices is 5-inducible, but not when every majority is carried by three voters against two, and we determine which obstructions survive deleting a vertex or reversing an arc. The method is a search that places one alternative at a time, always the one with the fewest options left, and exploits the symmetries of the tournament. Its negative results are certified by machine-checkable proofs, and all results are reproducible from a public repository.

Keywords: majority voting; tournaments; McGarvey's theorem; Paley tournaments; constraint search; SAT

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1. Introduction

Majority rule imposes no structure at all on collective preference when the electorate is unconstrained. For any pattern of pairwise majorities one cares to specify, cycles and all, some profile of voters produces exactly that pattern (McGarvey 1953); the Condorcet paradox is the smallest instance of this freedom rather than an exception to it. What makes the observation bite is that it needs *many* voters, and the question of how many is the question this paper is about.

Once the electorate is fixed, the freedom disappears. A panel of k voters admits only finitely many profiles, so it can produce only finitely many patterns of pairwise majority, and once the number of alternatives is large enough most patterns are beyond it. A pattern that some k-voter panel does produce we call *k-inducible*. The question is then one about small deliberative bodies: on an agenda of n alternatives, is a committee of k members expressive enough to realise every collective preference one could write down, or are there patterns of majority preference that k members can never produce, however they rank the alternatives?

For three voters the answer is known. Every pattern on at most seven alternatives is realisable, and on eight alternatives one is not, by the work of Shepardson and Tovey (2009) and of Eggermont, Hurkens and Woeginger (2013). We write $N(k)$ for the least number of alternatives at which some pattern escapes k voters, so that $N(3) = 8$. For five voters – the smallest committee size at which the question is open, and a common size for panels, benches and selection committees – the answer was known only to lie between 12 and 38, and no explicit pattern on fewer than 43 alternatives was known to escape five voters.

This paper narrows that to $13 \leq N(5) \leq 22$, and both ends are constructive. Five voters suffice on every agenda of at most twelve alternatives: we verify this for all $\approx 9.0 \times 10^8$ patterns on eleven alternatives

together with all of their one-alternative extensions. And on twenty-two alternatives there are patterns that five voters cannot produce: two tournaments, each obtained by removing one alternative from a doubly regular tournament on twenty-three. The first step on that way is the Paley tournament P_{23} on twenty-three alternatives – the instance a general-purpose SAT solver failed to decide in six weeks of computation in the work of Bachmeier et al. (2019), which our search settles in twenty-two hours on a laptop.

The pattern that first defeated five voters is worth describing in words, because it is not a pathology. P_{23} is as balanced as a collective preference on twenty-three alternatives can be: every alternative defeats exactly eleven others, and every pair of alternatives jointly defeats exactly the same number. It is the least decisive pattern available, in the sense that no alternative and no pair is distinguished from any other. That is precisely what a small panel cannot manufacture. A committee of five produces its majorities by accumulating agreement across five rankings, and the arithmetic of doing so leaves traces – some alternative better served than another, some pair more often aligned. Perfect balance admits no such traces. The obstruction to representation by a small electorate is therefore balance, not irregularity, which is the opposite of what a reader expecting hard instances to look messy would guess. Nor is it symmetry: P_{23} is also highly symmetric, but the other 36 doubly regular tournaments on twenty-three alternatives are just as balanced, 32 of them have no symmetry at all, and all of them escape five voters too (Section 3.7).

Two further findings sharpen the picture, and both concern *how* a majority is carried rather than whether it exists. A five-member panel can carry a pairwise majority by three votes to two, by four to one, or unanimously, and it is natural to ask whether the bare three-to-two margins suffice. They do not: P_{19} is realisable by five voters, but every profile realising it must carry some majority by more than the bare minimum (Section 3.4). And the failures are fragile in a precise sense: reversing a single pairwise majority in P_{19} or in P_{23} restores realisability (Sections 3.3 and 3.6). A pattern out of reach of five voters is thus not out of reach by a comfortable margin – it sits one pairwise comparison away from being realisable.

We follow the notation of our companion paper (Chindelevitch and Harutyunyan 2026) throughout, which we briefly restate here. Let T be a *tournament* on the vertex set V, $|V| = n$: a digraph in which every unordered pair $\{i, j\}$ carries exactly one of the arcs $i \to j$ or $j \to i$. A *k-profile* consists of k linear orders $\pi_1, \ldots, \pi_k$ on V. The *majority tournament* of such a profile puts an arc from i to j whenever $|\{v : i \prec_{\pi_v} j\}| > k/2$. For odd k this is well-defined on every pair, and the result is a tournament. We say T is *k-inducible* if some k-profile has T as its majority tournament, and we call such a profile a *witness* for T. The *converse* of T is the tournament obtained from T by reversing the direction of every arc.

We call the number of voters who rank i above j the *support* of the arc $e : i \to j$, and denote it $c(e)$. If $c(e) > k/2$, the *margin* of e is $2c(e) - k$. In a witness every arc's support exceeds $k/2$, so for $k = 5$ it is 3, 4 or 5. A witness has *margin at most M* if every arc has margin at most M. For $k = 5$ the possible margins are 1, which we call *unit margin*, 3 (every support in $\{3, 4\}$), and 5 (no restriction, i.e. plain majority). A unit-margin witness is in particular a majority witness. In Chindelevitch and Harutyunyan (2026) we conjectured that the margin-based hierarchy is strict, which for $k = 5$ asserts two separate things: that some tournament is 5-inducible with margin at most 3 but not with unit margin, and that some tournament is 5-inducible but not with margin at most 3. We settle the first here since P_{19} is such a tournament (Section 3.4), while the second remains open (Section 4.2).

One further quantity recurs below. Informally, the *predictability* $\alpha^*(T)$ is the largest fraction $\alpha > 1/2$ for which there is some profile, on any number t of voters, such that every arc of T is supported by at least αt of them; it is the value of a linear program over the linear orders of V. Since every arc of a k-voter witness is supported by more than $k/2$ of the voters, k-inducibility forces $\alpha^*(T) \geq (k + 1)/2k$, which is $\frac{2}{3}$ at $k = 3$ and $\frac{3}{5}$ at $k = 5$. The implication is only one-directional: α^* can meet the threshold and the tournament still fail to be inducible. Shepardson and Tovey (2009) introduced the notion; see Section 1.4 in Chindelevitch and Harutyunyan (2026) for the computations and the counterexamples at both $k = 3$ and $k = 5$.

Counting arguments show that almost all tournaments fail to be k-inducible for any fixed k, yet the *explicit* examples they yield are astronomically large. This paper closes much of the gap for $k = 5$. Let $N(k)$ be the least order of a tournament that is not k-inducible. The answer is only known exactly for $k = 3$. For $k = 5$ the state of the art before this work was

$$12 \leq N(5) \leq 38,$$

both ends of which are due to our companion paper (Chindelevitch and Harutyunyan 2026): the lower bound came from an exhaustive census of all $\approx 9.0 \times 10^8$ tournaments on 11 vertices, all of which admit a unit-margin five-voter profile, improving on the bound $N(5) \geq 11$ of Bachmeier et al. (2019).

The upper bounds in the literature are of two different kinds and they are not always comparable. Some are non-constructive, and obtained by counting: one compares the number of tournaments of a given order with the number that k linear orders can produce, and when the first exceeds the second, some tournament of that order must be missed. Bachmeier et al. (2019) argued this way to reach 41, and Chindelevitch and Harutyunyan (2026) sharpened the comparison by restricting it to regular and near-regular tournaments, reaching 38. Other bounds are constructive, and exhibit an explicit tournament. The record for $k = 5$ was P_z on $z = 603{,}979{,}799$ vertices (Bachmeier et al. 2019), from a dominating-set theorem of Alon et al. (2006) together with a construction of Graham and Spencer (1971), which we improved to P_{43} (Chindelevitch and Harutyunyan 2026), the first example of moderate order. Our bespoke algorithmic decision procedure shows that P_{23} is not 5-inducible, which improves both the best non-constructive (counting) as well as the best constructive (explicit) upper bounds to $N(5) \leq 23$, and then that two tournaments on 22 vertices are not 5-inducible either, so that $N(5) \leq 22$.

1.1 Special properties of Paley tournaments

The Paley tournament P_q, for a prime $q \equiv 3 \pmod 4$, has vertex set $\mathbb{F}_q$ and an arc from i to j whenever $j - i$ is a nonzero square, or *quadratic residue*; we write QR for the set of these, so that $i \to j$ exactly when $j - i \in \mathrm{QR}$. As everywhere in this paper, vertices are numbered from 1, vertex $i + 1$ being the field element i. It is *doubly regular*: every vertex has out-degree $(q - 1)/2$ and every ordered pair of vertices has exactly $(q - 3)/4$ common out-neighbours. Its automorphism group, written $\mathrm{Aut}(P_q)$ and formally defined in Section 2, has order $q(q - 1)/2$. This makes P_q *arc-transitive*, meaning that for every pair of arcs e, e' of P_q there is an automorphism of P_q that takes e into e'. It is also *self-converse*, meaning that it is isomorphic to its own converse; the relevant mapping is simply multiplication by -1, which is easily checked to not be an automorphism when $q \equiv 3 \pmod 4$.

1.2 Why Paley(23) is hard to classify and Paley(43) was easy

Paley tournaments are natural candidates for non-inducibility because they are far from being transitive, and because their symmetry makes an otherwise hopeless search merely very hard. Bachmeier et al. (2019) call them *quadratic residue tournaments* and write Q_q; their orientation is the converse of ours. Since reversing every voter's order reverses every arc of the majority tournament and changes no support, a tournament and its converse have the same *majority dimension*, the least k for which a tournament is k-inducible, and are inducible at exactly the same margins. We use the name Paley and the notation P_q throughout.

The SAT study of Bachmeier et al. (2019) establishes

$$\dim(Q_3) = \dim(Q_7) = 3, \qquad \dim(Q_{11}) = \dim(Q_{19}) = 5,$$

where dim denotes majority dimension, and then stops exactly where the problem becomes hard:

> "Unfortunately, we were not able to check whether the majority dimension of Q_{23} is equal to 5 or larger as the SAT solver did not terminate within a total of six weeks."

By contrast, our previous proof that P_{43} is not 5-inducible (Chindelevitch and Harutyunyan 2026) never searched the profile space in general. It rests on a quantity we call the *slack*,

$$\mathrm{slack}_5(T) = 5\,\mathrm{MAS}(T) - 3C, \qquad C = \binom{n}{2},$$

where MAS is the maximum number of arcs of T that a single linear order can agree with. Summing the majority condition over all arcs bounds how far the five voters can *collectively* fall short of that maximum. A small slack restricts the voters to be close to the best orders, and at $q = 43$, where $\mathrm{slack}_5 = 6$, that restriction allowed us to fully enumerate about 1.8 million cases.

While the same argument is valid at every q, the slack is *not* a monotone function of q (table from Chindelevitch and Harutyunyan 2026):

q	7	11	19	23	31	43
slack_5	7	10	22	**46**	30	**6**

Since P_{43} attains a locally minimal slack, while P_{23} attains a local maximum, the method of Chindelevitch and Harutyunyan (2026) is unusable at $q = 23$. For any q,

$$\text{slack}_5(P_q) \;=\; 5C\left(\alpha^*(P_q) - \tfrac{3}{5}\right),$$

where α^* is the predictability; equality holds because P_q is arc-transitive (Chindelevitch and Harutyunyan 2026). In particular, the necessary condition $\alpha^* \geq 3/5$ is far from binding, so no argument resting on the predictability can certify infeasibility.

Predictability separates $k = 5$ from $k = 3$. At $k = 3$ predictability decides exactly at $N(3)$: the 96 non-3-inducible tournaments on eight vertices have $\alpha^* = \frac{13}{20}$, which is *below* $\frac{2}{3}$, so the relaxation detects them, while all other tournaments on eight vertices or fewer with at least one cycle have $\alpha^* = 2/3$ (Shepardson and Tovey 2009), and $N(3) = 8$ follows. At $k = 5$ it does not: the table above has $\alpha^* > \frac{3}{5}$ at every q, yet P_{23}, P_{31} and P_{43} are all non-inducible, with P_{23} being the farthest *above* the threshold among the three.

1.3 Contributions

1. $N(5) \leq 22$ (Section 3.2). Two non-isomorphic tournaments on 22 vertices, each a doubly regular tournament on 23 vertices with one vertex deleted, are not 5-inducible, and both are vertex-critical. One of the two refutations is formally certified (Appendix B).
2. **P_{23} is not 5-inducible** (Section 3.1). This settles the instance on which the SAT solver of Bachmeier et al. (2019) did not terminate within a cumulative six weeks, and was the first step from 38 down to 23. We also decide P_{31}, which does not improve the bound but demonstrates the method's reach: the larger tournament is settled in less time than P_{23}.
3. **P_{19} is 5-inducible, but not 5-inducible with unit margin** (Section 3.4). This is an example suggesting that $k = 5$ may qualitatively differ from $k = 3$. For $k = 3$, the first k-inducible tournaments that are non-k-inducible with unit margin appear after $N(k)$, order 9 vs order 8 (Chindelevitch and Harutyunyan 2026). For $k = 5$ they appear no later than order 19, so the order is reversed unless $N(5) \leq 19$, which the search of Appendix G argues against.
4. **P_{19} and P_{23} are arc-critical** (Sections 3.3 and 3.6). Reversing an arc in P_{19} restores 5-inducibility at unit margin, and reversing an arc in P_{23} restores unrestricted 5-inducibility.
5. **P_{31} and P_{43} are not vertex-critical** (Section 3.5). Deleting a vertex from P_{31} or P_{43} leaves a tournament that is still not 5-inducible.
6. **Three formally certified negative results** (Appendix B). P_{19} is not 5-inducible at unit margin, and P_{23} and $\Delta_6 - 5$ are not 5-inducible. The search is split into independent subproblems (*cubes*); each is refuted by the SAT solver CaDiCaL, which emits its refutation as a machine-checkable *LRAT* proof; and every one of those proofs is then rechecked by a separate program that shares no code with the solver.
7. **Every tournament on 12 vertices is 5-inducible** (Appendix C). The analysis settles all 2,048 one-vertex extensions of each of the $\approx 9.0 \times 10^8$ tournaments on 11 vertices, which is complete because every 12-vertex tournament is such an extension of each of its twelve one-vertex deletions.

1.4 Related work

Two papers stand immediately behind this one, in different ways. Our companion paper (Chindelevitch and Harutyunyan 2026) is the prior art for the *bound*: it produced the first explicit non-5-inducible tournament of modest size, P_{43}, and its two bounds are what we measure against: $N(5) \leq 38$ by counting and $N(5) \leq 43$ explicitly. Bachmeier et al. (2019) is the prior art for the *problem and the method*: it brought SAT to bear on k-majority digraphs, characterised the 3-majority case building on Dushnik and Miller (1941), and established hardness results for voting with a constant number of voters.

The value of $N(3)$ is settled. Shepardson and Tovey (2009) introduced the predictability $\alpha^*(T)$ defined above. A 3-voter profile is exactly a $\frac{2}{3}$-supermajority representation, so $\alpha^*(T) < \frac{2}{3}$ rules out 3-inducibility; their 8-vertex tournament has $\alpha^* = \frac{13}{20}$, which gives $N(3) \leq 8$. They also prove that every tournament on at most 7 vertices has $\alpha^* \in \{\frac{2}{3}, 1\}$, and Eggermont, Hurkens and Woeginger (2013) then show by direct computation using an ILP formulation that every tournament on 7 vertices is 3-inducible, and every tournament on 8 or 9 vertices is 5-inducible, yielding $N(3) = 8$ and $N(5) \geq 10$.

Our certified computation (Appendix B.1) is standard. It splits the problem into independent subproblems, using the *cube-and-conquer* paradigm (Heule et al. 2012): a SAT instance is split by fixing a few variables at a time, giving many partial assignments called *cubes*, and each cube is handed to the solver as a problem of its own. We adopt the standard of evidence set by the Boolean Pythagorean triples proof (Heule et al. 2016; Cruz-Filipe et al. 2019), in which every subproblem emits a machine-checkable refutation that an independent checker validates. Our refutations are in the LRAT format (Cruz-Filipe et al. 2017) (Appendix B.1), a format in which the solver records each clause it derives so that an independent checker can replay and confirm the entire derivation.

This paper grew out of our companion paper (Chindelevitch and Harutyunyan 2026), whose constructive bound $N(5) \leq 43$ came from a structural argument, and whose counting bound $N(5) \leq 38$ names no explicit tournament. The question of what a genuine decision procedure would reach at smaller orders led to the search of Appendix A.

The rest of this paper is organised as follows. Section 2 sets out the notation, the symmetry reductions and the search in outline. Section 3 presents the results: P_{23} and P_{31} (Section 3.1), the two 22-vertex obstructions that give $N(5) \leq 22$ (Section 3.2), vertex- and arc-criticality (Sections 3.3 to 3.6), and other structured families (Section 3.7). Section 4 discusses where the method may transfer and what remains open, and Section 5 describes the reproducibility material. Appendix A describes the search engine and the cost of every computation, Appendix B the formal certificates, Appendix C the analysis of order 12, Appendix D the exact enumeration counts, Appendix E the construction for locally transitive tournaments, Appendix F criticality at 21 vertices, and Appendix G the search below 22 vertices.

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2. Preliminaries

Throughout, T is a tournament on $V = \{1, \ldots, n\}$ and $k = 5$ unless stated otherwise. We write $N^+(v)$ and $N^-(v)$ for the out- and in-neighbourhoods of v. A tournament is *regular* if $|N^+(v)| = |N^-(v)|$ for each $v \in V$; this can only happen for n odd. Following Chindelevitch and Harutyunyan (2026), D_n, R_n and S_n denote the numbers of isomorphism classes of, respectively, all tournaments, regular tournaments and self-converse tournaments on n vertices. The text gives them to two significant figures and Appendix D tabulates their exact values. For any $S \subseteq V$, the sub-tournament of T induced by S is the tournament obtained by restricting T to the vertices in S and the arcs between those vertices.

We call a tournament T *vertex-critical* for k-inducibility at margin M if T is not k-inducible at margin M, but $T - v$ is k-inducible at margin M for every vertex v. Similarly, we call T *arc-critical* for such a property if T has it, but any T' obtained from T by reversing an arc does not.

2.1 Symmetry reductions

An *automorphism* of a tournament T is a permutation of its vertices carrying every arc to an arc; these form a group $\mathrm{Aut}(T)$ under composition, and T is *rigid* when $\mathrm{Aut}(T)$ is trivial. For a permutation group Γ on a set X, the *orbit* of x is $\Gamma x = \{\gamma(x) : \gamma \in \Gamma\}$, and the orbits partition X. This group is *transitive* on X when this partition is trivial (equivalently, for any $x, x' \in X$ there is a $g \in \Gamma$ with $g(x) = x'$). We use orbits of $\mathrm{Aut}(T)$ on the vertices, on the arcs, and on the *ordered pairs* of distinct vertices, where γ sends (u, v) to $(\gamma(u), \gamma(v))$. An ordered pair is either an arc or the reverse of one, so that third action has at least two orbits.

Two symmetries act on witnesses and both are exploited by every method here: the $k!$ relabellings of the voters, and $\mathrm{Aut}(T)$ acting on the vertices. Each yields a reduction without loss of generality. As a witness can be transformed by an automorphism, we may in some cases decide in advance what some voter's top two vertices are.

Lemma 2.1 (orbit anchoring). *Let $\mathrm{Aut}(T)$ act on the ordered pairs of distinct vertices, and choose one representative pair from each orbit, say $(a_1, b_1), \ldots, (a_m, b_m)$. If T is k-inducible, then T has a witness in which, for at least one index i, some voter ranks a_i first and b_i second.*

Proof. Let $\pi_1, \ldots, \pi_k$ be a witness and let (u, w) be the top two vertices of π_1. The pair (u, w) lies in one of the orbits; let (a_i, b_i) be that orbit's representative, so there is $g \in \mathrm{Aut}(T)$ with $g(u) = a_i$ and $g(w) = b_i$. Applying g to every voter gives a profile whose majority tournament is $g(T) = T$, so it is again a witness, and its first voter ranks a_i first and b_i second. ■

Thus we may search only for witnesses whose top pair is one of the m chosen pairs. The lemma only guarantees that *some* i works, not any particular one.

For a Paley tournament $m = 2$, and the representatives can be exhibited without computing $\mathrm{Aut}(P_q)$. For $a \in \mathrm{QR}$ and any t, the map $x \mapsto a(x - t)$ multiplies every difference by a, which cannot change whether a difference lies in QR, so it is an automorphism; it sends the ordered pair (t, s) to $(0, a(s - t))$, and as a ranges over QR, $a(s - t)$ ranges over the coset $\mathrm{QR} \cdot (s - t)$. Since $\mathbb{F}_q^*$ is the disjoint union of QR and its complement, each of them one such coset, these maps alone leave only two orbits, separated by whether the difference lies in QR. The orbits of the full group are unions of these, so there are at most two, and since automorphisms carry arcs to arcs, at least two. They are therefore exactly the arcs and the non-arcs, and any arc together with any non-arc serves as representatives. On P_7, worked through in Figure A.1 of Appendix A.2, $\mathrm{QR} = \{1, 2, 4\}$, so the representatives are $(1, 2)$, whose difference 1 lies in QR, and $(1, 4)$, whose difference 3 does not; both lie inside the base $B = \{1, 2, 4\}$ used there, which is what lets the anchoring be checked from the base state alone, and panel (c) rejects an insertion that would lift a vertex above an anchored pair.

Corollary 2.2 (vertex anchoring). *Let $\mathrm{Aut}(T)$ act on the vertices and choose one representative $v_1, \ldots, v_r$ from each orbit. If T is k-inducible, then T has a witness in which some voter ranks some v_i first.*

Proof. The proof of Lemma 2.1 verbatim, with the action on vertices in place of the one on ordered pairs. ■

We say that a witness meeting either restriction is *anchored*. The two are alternatives rather than a sequence, each without loss of generality, and which is better depends on the tournament's number of vertex pair orbits vs vertex orbits, as well as the relative complexity of testing each restriction.

Lemma 2.1 must be given the full set of orbits. It licenses the restricted search *provided* the representatives meet every orbit. For every tournament we compute $\mathrm{Aut}(T)$ separately, with `nauty` 2.8.6 (McKay and Piperno 2014), and record its orbits alongside the run. Every automorphism group, orbit set and canonical form reported in this paper comes from that same tool.

Lemma 2.3 (voter lex-ordering). *If T is k-inducible, then T has a witness whose k orders are non-decreasing under any fixed total order on linear orders.*

Proof. The support $c(e)$ counts voters and is therefore invariant under permuting them, so any profile with the same multiset of orders induces the same tournament. Sorting the multiset gives the required witness. ■

The two lemmas may be applied simultaneously, which is not automatic: each is proved by transforming a

witness, and the second transformation must not undo the first. We apply Lemma 2.1 to a witness, then sort the resulting k orders by Lemma 2.3. Sorting only permutes the voters, while the property Lemma 2.1 guarantees, "*some* voter ranks a_i first and b_i second", depends on the multiset of orders and not on which voter carries it, so it survives the sort. Had Lemma 2.1 been sharpened to "voter 1 ranks a_i first and b_i second", the composition would fail. Both lemmas are therefore available to every formulation in Appendix A.

Together they reduce the running time substantially, in two different ways. Lemma 2.3 is absorbed into the definition of a base state: the k orderings of B are required to be non-decreasing, dividing the count by almost $k! = 120$ (almost, because a tuple with two equal orders is fixed by some of the $k!$ permutations).

Lemma 2.1 acts earlier still, *invalidating* base states outright, but only when *all* the representative pairs lie inside B. Suppose they do, and that a base state extends to an anchored witness, some voter of which ranks a_i first and b_i second overall. As $a_i, b_i \in B$, that voter ranks a_i first and b_i second *within* B, which the base state alone reveals. Contrapositively, a base state no voter of which tops B with a chosen pair extends to no anchored witness, hence to no witness at all, and may be discarded. If one representative has an endpoint outside B this filter does not apply; a base state failing the test on the representatives it sees may still extend to a witness anchored on the one it cannot. We call a base state *live* if it survives. On the five-vertex base we use on the Paley tournaments 2,591 of the 8,031 are live.

The search and the certification differ in *when* the condition in Lemma 2.1 is tested. Both test it at the start to determine the live base states. Our algorithmic approach keeps testing it during the search, since a base state fixes the orders only on B whereas the condition constrains the full orders: a voter picking (a_i, b_i) as its top pair in B ceases to be anchored once a later vertex is inserted above b_i. Pruning is sound once every representative has both endpoints placed, for then the test is monotone. An insertion can destroy an anchor but never create one, the placed vertices' relative order being already fixed, so a partial profile with no anchored voter has no extension with one. Pruning can only happen after all representatives' endpoints are placed. The SAT certification instead tests the condition once, at the root, where it decides which cubes reach the solver.

2.2 The search in five steps

The search extends base states. A base state fixes how the k voters rank B; the search inserts the remaining $n - |B|$ vertices into those rankings one at a time, backtracking as soon as the arcs already decided contradict T. A base state extended to k full rankings yields a witness; one that cannot be is eliminated; and T fails to be k-inducible exactly when every base state is eliminated. Algorithmically, with T and the margin regime as inputs, our approach has 5 steps:

1. **Pick a size.** Choose $b = |B|$, in practice 5 or 6.
2. **Pick a subtournament.** Choose B with $|B| = b$. Only $T|_B$ matters for what follows, so the choice can and should be optimised quickly (see Appendix A.4), since the base-state count varies by more than two orders of magnitude across the 12 isomorphism classes at $b = 5$.
3. **Find its base states.** Enumerate the k-tuples of linear orders of B whose supports agree with $T|_B$, counted up to voter permutation by Lemma 2.3.
4. **Determine the usable symmetries.** Compute $\mathrm{Aut}(T)$ with `nauty` and its orbits, then impose the ordered-pair anchoring of Lemma 2.1, the vertex anchoring of Corollary 2.2, or neither, discarding the base states no anchored witness can use.
5. **Run to completion, in parallel across base states.** The verdict is positive as soon as any base state yields a witness. It is negative only when *every* live base state has been searched to exhaustion.

Figure A.1 in Appendix A.2 shows all five steps on P_7 at $k = 3$. Steps 1–4 are fast and user-chosen; step 5 is a parallel computation. The same steps describe the SAT route, with step 5 replaced by handing each live base state to a CDCL solver as a *cube* (the partial assignment fixing the voters' rankings of B), so the two share their decomposition and their two lemmas and differ only in what refutes a leaf in the search tree.

3. Results

The two regimes we explore are *unrestricted* (or plain majority) inducibility, and *unit margin*. The former is the standard notion of inducibility, and the latter asks that every arc be carried by exactly three voters against two. Only an unrestricted refutation bounds $N(5)$. However, the lemma below shows that for the Paley family and most tournaments derived from it, the unrestricted inducibility is equivalent to inducibility with margin at most 3 (i.e. support at most 4).

Lemma (3-cycle bound). *Let $u \to v \to w \to u$ be a directed triangle of T and let $\pi_1, \dots, \pi_k$ be any linear orders with k odd. Then*

$$c(u \to v) + c(v \to w) + c(w \to u) \le 2k.$$

Consequently, in any majority witness, no arc lying in a directed triangle has unanimous support k.

Proof. A linear order cannot rank u above v, v above w and w above u simultaneously, so each voter agrees with at most two of the three arcs; summing over the k voters gives the inequality. If $c(u \to v) = k$, the other two supports sum to at most k, contradicting the majority requirement that each be at least $(k+1)/2$. □

For $k = 3$ the second conclusion is Proposition 1 of Milosz, Hamel and Pierrot (2018), with the same argument; we need it for every odd k, and in the summed form above.

3.1 Paley(23) and Paley(31) are not 5-inducible

Theorem. *P_{23} is not 5-inducible. Consequently $N(5) \le 23$.*

The proof is a search that runs to completion. The base is $\{1, 2, 3, 6, 12\}$, which induces the regular tournament on 5 vertices and gives 8,031 base states. They are reduced to 2,591 by Lemma 2.1, by requiring one voter to rank either $(1, 3)$ or $(1, 6)$ as the first pair. None of them produces a witness. This statement is also formally certified in Appendix B.1.

Read as a statement about voting, it says that a five-member panel cannot produce the collective preference P_{23} on twenty-three alternatives: no assignment of five rankings to five voters, out of the $23!^5$ available, has P_{23} as its pairwise-majority relation. It is the first such statement for an agenda of fewer than forty-three alternatives, and the first for any tournament whose non-realisability a general-purpose solver had been unable to settle.

The same method settles a larger case: P_{31} is not 5-inducible, over 21,009 base states with a different base, reduced to 4,007 by Lemma 2.1. Appendix A.5 shows how the runtime scales in q, and Appendix A.6 tabulates the cost of every computation in this paper. The value of settling P_{31} is to show that the bespoke approach decides 31-vertex instances on a single laptop within a day, something that SAT and ILP solvers are unable to do.

3.2 Two tournaments on 22 vertices that five voters cannot produce

P_{23} is one of 37 doubly regular tournaments on 23 vertices, which the catalogue of McKay (2026) lists up to isomorphism. We write Δ_i for the i-th tournament in that list, with its vertices numbered from 1 in the catalogue's own labelling, and $\Delta_i - v$ for the tournament left when vertex v is deleted. Every one of the 37 is beyond five voters, at unit margin and unrestricted alike (Section 3.7), so the doubly regular tournaments on 23 vertices are a natural place to look for a smaller obstruction one vertex down.

Theorem. *The tournaments $\Delta_6 - 5$ and $\Delta_{28} - 5$, on 22 vertices, are not 5-inducible. Consequently $N(5) \le 22$.*

The two are not isomorphic, and neither is isomorphic to the converse of the other, so they are two separate obstructions rather than one counted twice. Each has out-degrees 10 and 11, eleven vertices of each, and every arc lies in at least five directed triangles, so by the 3-cycle bound no arc of a witness could be unanimous. Neither has a non-trivial automorphism, so Lemma 2.1 and Corollary 2.2 do not apply and every base state has to be searched. Each proof is a complete search over all 8,031 base states of a five-vertex base inducing the regular tournament on five vertices, none of which extends to a witness. The two searches together took about 3,100 cluster core-hours (Appendix A.6). The refutation of $\Delta_6 - 5$ is also formally certified, by the same pipeline as those of P_{19} and P_{23} (Appendix B.1).

Read as a statement about committees, the theorem moves the threshold at which five voters stop being enough by one more alternative: on an agenda of twenty-two alternatives there are already patterns of pairwise majority that no five-member panel can produce. Both patterns sit one alternative away from perfect balance, and neither they nor the tournaments they come from have any symmetry: the only automorphism of Δ_6 and of Δ_{28} is the identity. What defeats a small electorate here is therefore balance, not symmetry.

Both obstructions are *vertex-critical*: deleting any one of the 22 vertices of either leaves a 5-inducible tournament, so neither yields a 21-vertex obstruction by a further deletion. Appendix G summarises the wider search below 22 vertices, which has found no obstruction there.

3.3 Criticality across the Paley family

Arc-criticality implies vertex-criticality: reversing an arc incident to v leaves $T - v$ unchanged, and the restriction of a witness to $V \setminus \{v\}$ is a witness for $T - v$. Each cell below therefore records only the *strongest* property established, at each of the two margins. "Not vertex-critical" therefore also rules out arc-criticality.

q	unit margin	unrestricted
19	**arc-critical**	5-inducible, so criticality does not apply
23	not vertex-critical	**arc-critical** (Section 3.6)
31	not vertex-critical	**not vertex-critical** (Section 3.5)
43	not vertex-critical	**not vertex-critical** (Section 3.5)

P_{19} is the only member critical at unit margin, and it is critical in the strongest sense: reversing any one arc makes it unit-margin inducible, which by the implication also gives vertex-criticality. At the other end, P_{31} and P_{43} fail to be vertex-critical at *both* margins. The interesting cell is $q = 23$: it is *vertex-critical unrestricted*, but *not* vertex-critical at unit margin. In other words, $P_{23} - v$ is 5-inducible, but not with unit margin, analogous to the following result.

The two 22-vertex obstructions of Section 3.2 are vertex-critical unrestricted; whether either is arc-critical is open.

3.4 Paley(19) is 5-inducible, but not with unit margin

That P_{19} is 5-inducible is due to Bachmeier et al. (2019), who computed $\dim(Q_{19}) = 5$. Our search recovers it independently and contributes the negative half: it is *not* 5-inducible with unit margin. This is a complete refutation over all 2,200 base states of the base $\{1, 2, 3, 4, 6\}$, and it is the second statement we certify formally in Appendix B.1. These two facts supply the separation in Section 1: no arc of P_{19} can be unanimous, by the 3-cycle bound above, so it is 5-inducible with margin at most 3 and not with unit margin, so the inclusion of the unit-margin regime in margin at most 3 is strict.

For a panel, this is the difference between producing a collective preference and producing it comfortably. P_{19} is within reach of five voters, but not by bare three-to-two majorities alone: any five rankings that realise it must settle at least one pair by four votes to one. Strength of majority is thus a resource in its own right, separate from the number of voters, and the two do not substitute freely for one another.

Our three certified refutations, P_{23} and $\Delta_6 - 5$ at unrestricted margin in Sections 3.1 and 3.2 and P_{19} at unit margin, are established formally by cube-and-conquer computations in which every leaf is refuted by CaDiCaL 2.0.0 (Biere et al. 2020; Pollitt et al. 2023) in LRAT and rechecked by `lrat-trim` (Pollitt et al. 2023), a separate program that shares no code with the solver, with the solver's own proof checking deliberately disabled. All three are complete on *both* halves (the leaves and the claim that the leaves are exhaustive). Those for P_{19} and P_{23} reduce to the same two human-checked lemmas, 2.1 and 2.3; that for $\Delta_6 - 5$, which has no non-trivial automorphism, needs only Lemma 2.3 (see Appendix B).

3.5 Paley(31) and Paley(43) are not vertex-critical

Vertex-criticality is the natural minimality notion for an *obstruction*, a tournament that is not 5-inducible, and P_{43}, the first explicit non-5-inducible tournament (Chindelevitch and Harutyunyan 2026), is the natural candidate.

Theorem. *Neither P_{31} nor P_{43} is vertex-critical for 5-inducibility.*

The orbit break. For any $q \equiv 3 \mod 4$, $\mathrm{Aut}(P_q - v)$ is *exactly* the stabiliser of v in $\mathrm{Aut}(P_q)$. One inclusion is immediate; for the other, deleting v removes an out-arc from every vertex of $N^-(v)$ and from no vertex of $N^+(v)$, so $P_q - v$ has just two out-degrees, $(q-1)/2$ on $N^+(v)$ and $(q-1)/2 - 1$ on $N^-(v)$, and an automorphism preserves out-degree, hence those two sets, hence extends to P_q by fixing v. Translating so that $v = 0$, that stabiliser is $\{x \mapsto ax : a \in \mathrm{QR}\}$, cyclic of order $(q-1)/2$, and it splits the remaining $q-1$ vertices into exactly two orbits of size $(q-1)/2$, namely QR and its complement. Two representatives, one from each, therefore meet every orbit, and the vertex anchoring of Corollary 2.2 applies with those two.

Each sweep runs over all 8,031 base states of a five-vertex base inducing the regular tournament on five vertices: $\{1, 2, 3, 4, 13\}$ in $P_{31} - v$ and $\{1, 2, 3, 4, 11\}$ in $P_{43} - v$. The two counts coincide because, as in Appendix A.1, the number of base states depends only on the induced subtournament and on the margin, and not on n.

Every arc of $P_q - v$ lies in at least $(q-3)/4$ directed triangles, so no arc of a majority witness could be unanimous, and margin ≤ 3 is not a restriction at all. The refutation at margin ≤ 3 is therefore an unrestricted refutation. We use the same approach for both $q = 31$ and $q = 43$.
We thus independently reconfirm the main result of Chindelevitch and Harutyunyan (2026), that P_{43} is not 5-inducible: if it were, so would be $P_{43} - v$, by deleting v from every voter's order.

3.6 Vertex and arc criticality for Paley(23)

Theorem. *P_{23} is arc-critical: reversing any single arc yields a 5-inducible tournament. Consequently it is vertex-critical.*

Let T^e denote P_{23} with the arc $(1, 2)$ reversed; since $\mathrm{Aut}(P_{23})$ is transitive on arcs, any two single-arc reversals of P_{23} are isomorphic, so we may pick $(1, 2)$ without loss of generality. The computation is a scan of the 8,031 base states of T^e, but now T^e is rigid, so Lemma 2.1 and Corollary 2.2 do not apply. The witness we found was verified independently of the search that produced it, by two checks on the five orders: each is a permutation of the 23 vertices, and the majority tournament they induce agrees with T^e in every one of the 253 arcs and therefore differs from P_{23} in exactly e.

Vertex- and arc-criticality come apart, as far as they can. Arc-criticality implies vertex-criticality, so it is natural to ask whether the two coincide. They do not. Among the 110 vertex-transitive tournaments on 21 vertices, four are unit-margin obstructions, and all four are vertex-critical. However, arc-criticality takes all three of its possible values among those same four: two of them have every arc critical, one has exactly half, and one has none at all, so no statement of the form "this tournament is or is not arc-critical" captures the situation (full details in Appendix F).

3.7 Other structured tournament families

Beyond the Paley family we swept the structured families catalogued by McKay (2026) exhaustively for $n < 23$. Every verdict is negative for the bound, so we only report them briefly.

First, *the vertex-transitive family contains no majority obstruction at all for $n < 23$*: all 110 are 5-inducible, 106 of them already at unit margin and the remaining four by exhibited majority witnesses. Since only a majority refutation can move $N(5)$, that family is closed as a source of candidates below 23, and the closure survives one-arc perturbation, since reversing any single arc of the four unit-margin obstructions, one representative per arc orbit, again leaves a majority-inducible tournament in all 40 cases.

Second, *both* doubly regular tournaments on 19 vertices, P_{19} and the one other of that order, are unit-margin obstructions and both are majority-inducible, so the separation that makes P_{19} interesting is not a property

of P_{19} in particular. Nor is its criticality: the second one is *arc-critical* at unit margin too. Its automorphism group has order 3, so its 171 arcs fall into 57 orbits of size 3, and reversing the representative of any one of them yields a tournament with a unit-margin witness: 57 orbits and 57 witnesses. Arc-criticality at unit margin is therefore shared by both doubly regular tournaments of order 19.

Third, every regular tournament on at most 15 vertices is inducible at unit margin, over all $R_{15} \approx 1.8 \times 10^{10}$ of those on 15 vertices, and with the instance counts summing to OEIS A096368(7) (OEIS Foundation Inc. 2026c). Likewise, every self-converse tournament on 13 vertices is inducible at unit margin, over all $S_{13} \approx 9.5 \times 10^7$ of them. With the order-11 census of Chindelevitch and Harutyunyan (2026) and the order-12 analysis of Appendix C, that settles every self-converse tournament on at most 13 vertices.

Fourth, the order 23 is complete for both families. Every vertex-transitive tournament on 23 vertices other than P_{23} is 5-inducible, while none of the 37 doubly regular ones is, at unit margin or unrestricted. Since reversing every arc preserves inducibility, the 37 reduce to 19 classes up to taking the converse, and each was settled by a complete search; 16 of them ran on the cluster, at about 26,500 cluster core-hours together.

Taken together with the Paley results above, these sweeps turn up seventeen tournaments that five voters can produce, but not with unit margin: 2 doubly regular ones at order 19, 4 vertex-transitive ones at order 21, and 11 circulant ones at order 23.

One family needs no search. A tournament is *locally transitive* if every in- and out-neighbourhood induces a transitive subtournament. Every such tournament is k-inducible at unit margin for every odd $k \geq 3$, by an explicit construction rather than a computation; Appendix E provides it. This is the one structural family for which the inducibility question is settled outright.

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4. Discussion and open problems

4.1 Other applications of incremental depth-first search with MRV

Subject to the caveats of Appendix A.6, the shape of the engine may transfer: a placement search in which each decision *refines*, rather than recomputes, the candidate sets of all remaining objects, with a dynamic minimum-remaining-values (MRV) order and an externally discharged symmetry break. The conditions are specific: the extensions of a partial solution must factor as a product that shrinks monotonically under each decision, and the symmetry group must be computable; a general-purpose solver may win if they do not apply.

Neither device is new in isolation, and the closest antecedent is Knuth's Algorithm X with dancing links (Knuth 2000). Its column-choice rule (take the column with the fewest remaining ones) is our MRV order, and its link surgery does what our refinement does: the state is altered in place and put back on backtrack, so a node costs what changed rather than what the whole state contains. The constraint is what differs. Exact cover partitions, whereas an arc of a tournament imposes a threshold, at least three of five voters, so the search here is not an exact cover instance and dancing links does not apply to it as it stands.

Two design lessons do seem to generalise. First, the incremental refinement, not the heuristic order, carried most of the speedup here, a factor of 110 against a factor of a few. Second, a symmetry break should be *parameterised* by its representatives rather than hard-coded, so that its correctness obligation becomes a separate computation instead of an assumption buried in the search; Section 3.5 is a case where discharging that obligation required an explicit orbit calculation that the engine could not have performed for itself.

4.2 Conjectures and open problems

Question 1: Is $N(5) = 22$, that is, is every tournament on 21 vertices 5-inducible?

Both ends of $13 \leq N(5) \leq 22$ are constructive and improved from Chindelevitch and Harutyunyan (2026). The unit-margin obstructions at $n = 19$ and 21 cannot move the upper one, since every one of them is majority-inducible. The evidence below 22 comes from the descendants of the doubly regular tournaments

on 23 vertices (Appendix G): every tournament on 20 vertices obtained from one of them by deleting three vertices is 5-inducible, and so is every one on 21 vertices below either 22-vertex obstruction.

Conjecture 2: P_q is not 5-inducible for any eligible prime $q \geq 23$, i.e. $q \equiv 3 \pmod 4$.

It holds at $q = 23$, 31 and 43, and only finitely many q are in doubt: $\alpha^*(P_q) = \mathrm{MAS}(P_q)/C$ tends to $\frac{1}{2}$, so the predictability eventually falls below $\frac{3}{5}$, past which the necessary condition of Section 1.2 refutes 5-inducibility with no search at all. What is missing is a bound on where that range ends; the values in Section 1.2 fall too irregularly to extrapolate one.

Open problem 1: Vertex- and arc-criticality. Whether P_q minus *two* vertices is 5-inducible, equivalently whether P_q minus one vertex is vertex-critical, is open at $q = 31$ and $q = 43$.

Open problem 2: The margin hierarchy. The first of unit margin $\subseteq$ margin $\leq 3 \subseteq$ unrestricted is strict (Sections 3.4 and 3.7); whether the second ever is, that is, whether some tournament is inducible by five voters but not at margin ≤ 3, is open. The 3-cycle lemma forces such a tournament to have an arc in *no* directed triangle, which rules out every regular tournament. The conjecture of Chindelevitch and Harutyunyan (2026) is that the inclusion is strict at every odd $k > 3$.

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5. Reproducibility

5.1 The repository

The code, data and verdict files are at `https://github.com/Leonardini/TournamentsBeyond5Voters`. Its `CLAIMS.md` carries one row per statement made here, naming the artifact that holds the verdict and the command that re-derives it, and it ends with an explicit list of what the package does not establish. A permanent archive of the commit corresponding to the accepted version will be deposited on acceptance and its DOI recorded here.

The repository contains the search engine as a single C translation unit, the SAT encoding and cube tooling in Python, the certification driver, the independent witness verifiers, the tournaments as bit strings, and a verdict record for every result reported here. Each verdict records the exact command line, the base and break used, the coverage audit, the runtime, and the two root hashes for the formally certified results.

5.2 What must and what may not replicate

The certified computations publish two root hashes, and only one of them is portable.

ROOT (CNF) commits to the SHA-256 of every cube's CNF, in cube order. These files are generated by the repository's own code from the tournament and the base, and are reproducible on any machine: a third party who regenerates the cube set *must* obtain this exact value. It is the meaningful cross-check, because it proves they solved the same problems.

ROOT (proofs) additionally commits to the LRAT proof bytes. These are *not* portable, as they depend on the exact SAT solver and version used. A mismatch here is not evidence of an error.

We have run that check: on the cluster, under a different compiler and CaDiCaL build, both refutations gave matching roots at 70.9 and 567.6 cluster core-hours against the 25.1 and 236.0 of Appendix A.6, and both coverage instances rebuilt to their committed CNF hashes and were refuted again.

Search runtimes: nodes replicate, hours do not. A three-way version of that distinction applies to the uncertified sweeps. *Verdicts* replicate under any correct implementation, which is the claim the paper makes. *Node counts* replicate exactly, but only with five choices held fixed:

1. the same tie-breaking rule in the variable order and in the domain refinement; ties are common and a different rule explores a different tree of the same size class but not the same size;
2. the same vertex numbering of the tournament;
3. the same base size and base;

4. the same resulting set of base states, which follows from (2) and (3);
5. the same margin regime.

The engine prints all of these except (2) in a header on every run (margin, variable order, tie-break mode, symmetry break, base size, base, base arc pattern and base-state count), so a run log is a self-describing fingerprint. Condition (2) is the tournament file itself, which is why the repository stores tournaments rather than regenerating them from their constructions. *Core-hours* replicate only in order of magnitude; Appendix A.5 measures a factor of 2 to 5 per core between the two machines we used.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

[**Leonid confirmed; TO CONFIRM with Ararat.**] **Leonid Chindelevitch:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review and editing. **Ararat Harutyunyan:** Conceptualization, Formal analysis, Writing – review and editing.

Data availability

The code, data and verdict files supporting the findings of this study are at `https://github.com/Leonardini/TournamentsBey` described in Section 5.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) for exploratory reasoning, implementation assistance, and drafting and editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. The search engine

Both integer programming and satisfiability are central to this project rather than foils for it. Every computational claim in the companion paper (Chindelevitch and Harutyunyan 2026) is certified by an integer or linear program, and the integer program of A.1 below is that paper's formulation B.1, verbatim; the only machine-checked theorem in this paper is a SAT computation from end to end, with CaDiCaL (Biere et al. 2020) emitting LRAT proofs that `lrat-trim` checks. The claim here is narrow: on this one decision problem, at this one range of sizes, neither tool reaches the answer while a search built around the problem's own structure does, and the same SAT machinery is then indispensable for turning that answer into a proof.

A.1 Base states

Every search starts by fixing how the voters rank a handful of vertices and then looks for ways to add the remaining ones. We choose a *base* $B \subseteq V$ of b vertices (we use $b = 5$ or $b = 6$), and write $T|_B$ for the sub-tournament induced on B. When B is a base, a *base state of* B (we omit B when the context makes it clear) is a k-voter profile over B that is consistent with $T|_B$, i.e. k orderings of B whose majorities reproduce $T|_B$ at the margin being asked for. Base states differing only by a relabelling of the voters are counted once (see Lemma 2.3).

Each base state of B defines a self-contained subproblem: the rankings of B are given, and the question is whether the remaining $n - |B|$ vertices can be inserted so that every arc of T has the right support, so the search splits into independent components across base states. How many there are depends only on $|B|$, on $T|_B$ and on the margin regime, but *not* on n; thus, with the five-vertex bases we use on the Paley tournaments, which induce regular sub-tournaments, there are 8,031 of them in both P_{23} and P_{47}.

A.2 Depth-first placement search

The search maintains a partial profile: a set $S \subseteq V$ of *placed* vertices together with, for each voter, a linear order of S. It begins from a base state, which places B, and extends by inserting one unplaced vertex at a time. Inserting v means choosing, for each voter independently, a slot among the $|S| + 1$ available; the *domain* $D(v \mid S)$ is the set of k-tuples of slots consistent with every arc between v and S. A partial profile is extendable only if every unplaced vertex has a non-empty domain, and domains shrink monotonically as S grows, reducing the search to a constraint-propagation problem. Two decisions in the pseudocode below enable its efficiency (see Appendix A.4 for more details). First, the search always inserts an unplaced vertex of currently smallest domain (the minimum-remaining-values, or MRV, rule), the domain sizes being recomputed at every node rather than fixed once per base state; second, a child's domains are refined from its parent's.

```
ALGORITHM 1.  Placement search for k-inducibility of T

Input :  tournament T on vertex set V;  number of voters k; margin regime (majority or unit)
Output:  a witness profile, or a refutation

MAIN
  for each base state  (a lex-canonical assignment of the base B to k orders):
      S <- B
      D(v | S) <- fresh enumeration, for every v not in S
      if any D(v | S) is empty:  continue     // this base state is dead
      if DFS(S) succeeds:        return its witness
  return "not k-inducible"                    // no base state extends

DFS(S)
  if S = V:  return the profile               // all vertices placed
  v* <- argmin  |D(v | S)|   over v not in S  // MRV, recomputed at EVERY node
```

```
      ties broken toward  max | |out(v) & S| - |in(v) & S| |
  for each slot tuple p in D(v* | S):
      insert v* at p in each of the k orders;  S' <- S u {v*}
      for each w not in S':
          D(w | S') <- REFINE( D(w | S), p, w, v* )
          if D(w | S') is empty:  undo;  continue with the next p
      if DFS(S') succeeds:  return its witness
      undo                                    // arena pointer reset
  return failure

REFINE(D, q, w, u)                            // q = the slot tuple just used for u
  D' <- {}
  for each p in D:
      lift p to p' voter by voter:
          p_i <  q_i  ->  p'_i = p_i          // w stays before u
          p_i >  q_i  ->  p'_i = p_i + 1      // w stays after u, shifted by the insertion
          p_i == q_i  ->  BOTH are legal, so the tuple splits in two
      keep those lifts whose support  c_w(u) = #{ i : w before u }
      meets the requirement of the single new arc between w and u
  return D'
```

A.3 Integer programming and SAT are both too slow

The natural integer program has one binary $x_{v,ij}$ per voter v and pair $i < j$, meaning that v ranks i above j, with $x_{v,ji}$ read as $1 - x_{v,ij}$ so that antisymmetry is structural. Each voter is forced to be a linear order by forbidding both orientations of every 3-cycle,

$$0 \;\le\; x_{v,ij} + x_{v,jl} - x_{v,il} \;\le\; 1 \qquad (i < j < l),$$

and the profile is forced to induce T by one majority row per arc, $\sum_v x_{v,ij} \ge (k+1)/2$. This is a pure feasibility problem. At $q = 23$, $k = 5$ it has 1,265 binaries, 17,710 transitivity rows and 253 majority rows.

The obstruction is structural rather than a deficiency of the solver. Let us set $x_{v,ij} = 3/5$ on every arc of T and $2/5$ on every non-arc, the same for all voters. Every cyclic triple then reads $3/5 + 3/5 - 2/5 = 4/5$ and every transitive one $3/5$, both interior to $[0, 1]$, while every majority row is tight at exactly 3. The relaxation is therefore feasible, and since the problem carries no objective, LP feasibility yields no bound whatever: branch-and-bound is left with nothing to prune by except infeasibility discovered deep in the tree. That fractional point is also invariant under both symmetry groups acting on the problem (the $k!$ voter relabellings and $\mathrm{Aut}(T)$), so it supplies no branching direction either, and orbital branching recovers only the $k!$ factor. CPLEX and Gurobi both failed to close instances at $n = 19$ within a day.

The satisfiability encoding is *dissent-Boolean*: one variable per voter v and arc $i \to j$ of T, true when v ranks j above i and so dissents from that arc. Each arc bounds its dissents, at most $(k-1)/2$ of them for majority and exactly $(k-1)/2$ for unit margin, and each voter's dissent set is forced to leave a linear order triple by triple: a tournament is transitive exactly when it has no directed 3-cycle, so it is enough to forbid, for every voter and every triple, the two dissent patterns that would make that triple cyclic. For $k = 5$ that is $5\binom{n}{2}$ variables, 15 clauses per arc at unit margin and 10 per triple. It is compact and propagates well locally, and it is what both certified refutations of Appendix B are built on.

Table 4 of Bachmeier et al. (2019) reports their `Sat-Check-k-Majority` on uniform random tournaments, tabulating only entries whose average stays under 20 seconds. The $k = 5$ column reads

n	18	19	20	21	22	23	24
seconds	0.23	0.35	0.54	5.87	11.07	18.95	—

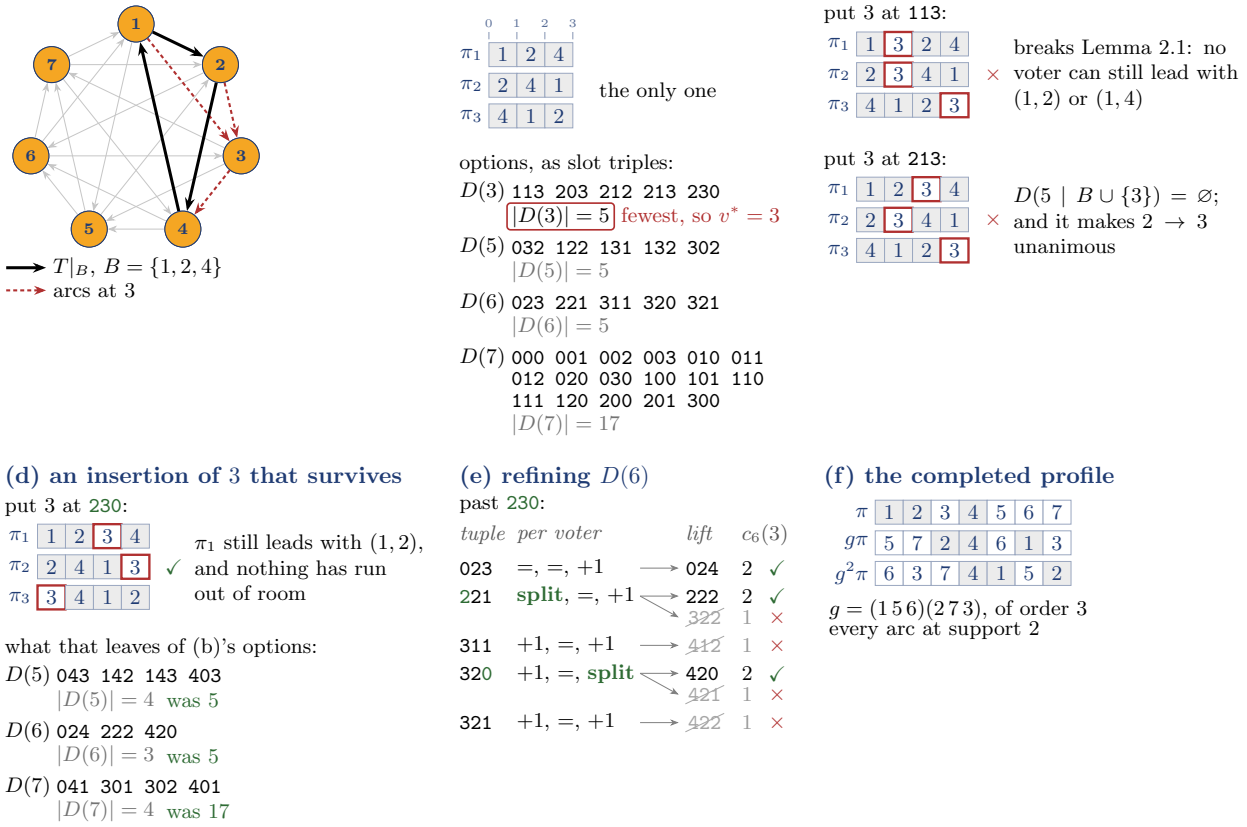


Figure A.1: Algorithm 1 on P_7 at $k = 3$, which is 3-inducible, so the run ends in a witness. A *slot triple* is one gap per voter, of the $4^3 = 64$ available, and a vertex's domain is those triples its arcs to B admit. Here the base is $B = \mathrm{QR}$, the squares modulo 7, and Lemma 2.1 guarantees that at least one voter must rank 1 first and either 2 or 4 second. Ranking 4 second forces the other two voters to 241, the only order getting both $4 \to 1$ and $2 \to 4$ right, and $1 \to 2$ is then left with support 1; so the base state in (b) is the only one. **(a)** The base $T|_B$ on $B = \{1, 2, 4\}$ in bold; dashed, the arcs at the first vertex inserted. **(b)** Steps 2–4: the base state is forced, and MRV takes the smallest domain. **(c)** Two insertions the search rejects, one by Lemma 2.1 and one by leaving 5 no slot at all. **(d)** One that survives, and what it leaves of (b). **(e)** REFINE lifts each parent tuple past the slot just used (unchanged below it, shifted above it, split where the two coincide, which is what green marks), and the one new arc then keeps or kills each lift. **(f)** The witness, every arc at support exactly 2. Without Lemma 2.1 the search returns all 7 witnesses, a single orbit of $\mathrm{Aut}(P_7)$, of order 21, so the profile is unique up to symmetry, as Shepardson and Tovey (2009) report; π generates the other two under the g shown, the map $x \mapsto 2x + 4$ of $\mathbb{F}_7$.

so on *random* instances the method leaves the table between $n = 23$ and $n = 24$, roughly doubling per added vertex. Random tournaments are the easy case: no automorphisms to rediscover and, being far from extremal, many witnesses. On the structured instance of the same order, P_{23}, the same solver did not terminate within a cumulative six weeks. The gap between 18.95 seconds and six weeks at equal n is the cost of symmetry.

A better encoding and an explicit symmetry break both help substantially and are still not enough (results not shown). What does work is supplying the solver with the decomposition itself: cube-and-conquer over base states, which is how both refutations are certified in Appendix B. A CDCL solver must otherwise discover that decomposition for itself, and cannot exploit $\mathrm{Aut}(T)$ at all without an explicit break in the encoding.

A.4 What makes the search fast

Four choices make the search of Appendix A.2 fast, and the second is the one that made $n = 19$ and $n = 23$ reasonable to compute.

A dynamic variable order. At every node we insert the unplaced vertex minimising $|D(v \mid S)|$, breaking ties toward the vertex whose arcs to S are most imbalanced. The important part is that the order is recomputed at every node rather than fixed per base state: on Paley instances a static order is not competitive, because how constrained a vertex is depends strongly on which part of the tournament has already been placed.

Incremental domain refinement. Recomputing $D(v \mid S)$ from scratch at each node is the dominant cost of a naive implementation; instead the child's domain is derived from the parent's by the three cases in `REFINE` above, after which the single new arc filters the lifted tuples by their support. Child domains are therefore refinements of parent domains, computed in time proportional to the surviving tuples rather than to the whole slot space, and domains live in a stack-disciplined arena indexed by (depth, vertex) so that backtracking is a pointer reset. Like for like this is worth a factor of about 110 in wall time.

Symmetry breaking. The $k!$ voter symmetry is absorbed exactly by the base-state definition, which counts non-decreasing tuples only. For $\mathrm{Aut}(T)$ the engine imposes the anchoring of Lemma 2.1 or of Corollary 2.2, whichever is stronger on the tournament: the condition is invariant under permuting voters, so it composes with the base lex-order, and monotone under insertion, so it can be tested during the search rather than only at the leaves.

Early termination and caching by arc pattern. A quick per-voter agreement bound, obtained by counting the agreement between voter i and the arcs between v and S two ways, proves $D(v \mid S)$ empty in $O(k|S|)$ time without enumerating it. And the per-base-state work is keyed by the *arc pattern* of $T|_B$ rather than by B itself, so base-state counts, which depend only on that pattern, are reused across every copy of the same base class and across every q.

Choosing the base. The base set is not a free parameter, and the fewest live base states is not the fastest choice: at $q = 31$ the class with 4,007 live states beat the class with 2,537, whose states are individually slower, so we rank the twelve 5-vertex classes by the product of the live count and a sampled time per state rather than by the count alone.

A.5 How the runtime scales in q

A note on $q = 27$. The series below includes $q = 27$, the one prime power among the orders we computed with. Since $27 = 3^3$, P_{27} must be built over $\mathbb{F}_{27}$ and not over $\mathbb{Z}/27$. P_{27} is the unique most symmetric doubly regular tournament on 27 vertices, with $|\mathrm{Aut}(P_{27})| = 1053 = 3q(q-1)/2$, because the field automorphisms contribute (in general $|\mathrm{Aut}(P_q)| = q(q-1)e/2$ when $q = p^e$). It is still transitive on arcs and on non-arcs, which is what Lemma 2.1 needs. We ran $q = 27$ to completion and it is not 5-inducible; since that does not bear on $N(5)$ we report it here only as a runtime datapoint.

On a fixed base decomposition the total runtime *decreases* as the tournament grows, and so does the node count: a larger Paley tournament is *more* constrained, so the search is refuted nearer the root and the tree above each base state is smaller rather than larger. The work per node moves the other way, rising monotonically from 10.8 to 57.4 microseconds across the four runs below, because a larger tournament carries larger domains and each refinement step costs more. The node count falls faster than the per-node cost rises, which is what makes the total fall. The full runs:

q	live base states	core-hours	nodes	seconds per live base state
23	2,591	34.03	1.14×10^{10}	47.3
27	2,537	31.04	6.22×10^{9}	44.0
31	4,007	26.89	3.43×10^{9}	24.2
43	2,537	22.87	1.43×10^{9}	32.5

P_{43} is decided in *less time than* P_{27}, and explores 4.3 times fewer nodes to do it. The last column is the third

divided by the second, so only the core-hours and the live base states are measured. Just two of these four runs share a decomposition, $q = 27$ and $q = 43$, on the same five-vertex base class and the same 2,537 live base states; $q = 23$ and $q = 31$ each used their own best configuration, and $q = 31$'s is the cheaper class of Appendix A.2, which is why its per-state figure is the lowest of the four. The core-hours fall monotonically in q either way. By contrast, P_{43} minus a vertex takes 185.5 core-hours, far more than P_{43} itself, as vertex deletion shrinks the automorphism group.

Additionally, the search gets faster as the margin is tightened. On the identical tournament ($P_{43} - v$), engine and 5-vertex base class, there is a factor of about 90. Requiring every arc to be at unit margin admits 2,200 base states where majority admits 8,031; additionally, under exact margin the last voter is *forced*, since the support accumulated over the first $k - 1$ voters determines what the last must contribute, so a whole level of the search collapses.

Two machines, and why their runtimes are not comparable. Some computations ran on a cluster rather than the laptop. Measured across six independent experiments, the cluster ran at 2 to 5 times the laptop's runtime per unit of work.

We are not claiming that bespoke search beats ILP or SAT solvers in general. Our engine is narrow: it exploits the specific facts that a partial profile's extensions form a product of per-voter slot choices, that this product refines monotonically, and that $\mathrm{Aut}(T)$ is large. For the specific Paley instances in this paper that the general-purpose methods did not settle for us, the search settles in hours on a single laptop.

A.6 Runtimes for every computation

All runs in this manuscript were performed on a single laptop (Apple M-series, 14 cores, 24 GB RAM), using at most 11 cores concurrently, with the following exceptions, which used a compute cluster: the census of regular tournaments on 15 vertices, the self-converse census of order 13, the order-12 census of Appendix C, the two 22-vertex sweeps of Section 3.2, 16 of the 19 unrestricted sweeps of doubly regular tournaments on 23 vertices in Section 3.7, and most of the search of Appendix G. Runtimes are reported in *core-hours*, the workers' running times summed; cluster core-hours are marked as such, and are not comparable with laptop ones (Appendix A.5).

computation	margin	verdict	core-hours
P_{19}, certified refutation	unit	not inducible	25.1
P_{23}, certified refutation	unrestricted	not inducible	236.0
$\Delta_6 - 5$, certified refutation	unrestricted	not inducible	1,292.7
$\Delta_6 - 5$, sweep (cluster)	unrestricted	not inducible	1,551.2
$\Delta_{28} - 5$, sweep (cluster)	unrestricted	not inducible	1,537.7
$P_{31} - v$, sweep	unrestricted	not inducible	239.5
P_{23}, one arc reversed	unrestricted	inducible	210.4
second doubly regular 19, all 57 arc reversals	unit	all inducible	3.59
regular tournaments on 15 vertices (cluster)	unit	all inducible	3,106
self-converse tournaments on 13 vertices (cluster)	unit	all inducible	1,310
all tournaments on 12 vertices (cluster)	unit	all inducible	$\approx$ 10,000

The four full Paley sweeps, $q = 23$, 27, 31 and 43, are tabulated with their node counts in Appendix A.5.

Appendix B. The formal certificates

B.1 The certified refutations

All three machine-checked results follow an established design rather than a new one. *Cube-and-conquer* is due to Heule, Kullmann, Wieringa and Biere (2012): split a hard instance into many partial assignments (*cubes*), hand each to a CDCL solver, and let the splitting supply the global structure the solver cannot find for itself. The standard of *proof* is the one Heule, Kullmann and Marek set for the Boolean Pythagorean

triples problem (Heule et al. 2016; Cruz-Filipe et al. 2019), and Schur number five (Heule 2018) and the Keller conjecture (Brakensiek et al. 2020) after it: every subproblem emits a machine-checkable refutation, validated by an independent checker rather than trusted. We claim no methodological novelty.

What an LRAT proof is. A solver's verdict of unsatisfiability is not self-evidently checkable, so the solver writes a *proof log*: a file recording, in order, the clauses it derives, each of which must follow from the input formula together with the clauses already recorded, the last being the empty clause. A separate checker replays that file and confirms every step, so what one trusts is the checker rather than the solver. The available formats record essentially this same derivation and differ mainly in how much work the replay costs. We use LRAT (Cruz-Filipe et al. 2017), the format CaDiCaL emits natively and `lrat-trim` consumes, in which checking is a single pass over the file.

Two differences are consequences of our setting. Our cubes are *base states*, a combinatorial decomposition, which lets the coverage of the cube set be certified separately (B.2) instead of resting on a splitting heuristic. And where Heule et al. (2016) published a cube certificate from which its refutations can be regenerated, we *verify and discard*: each LRAT proof is checked, hashed and deleted, and we only publish the hash value. The choice suits the scale. Verifying the Boolean Pythagorean triples proof costs about 13,000 core-hours to regenerate the refutations from the published cube certificate (Heule 2016), and about 16,000 to check them (Heule et al. 2016). Our certificates are orders of magnitude smaller: 25.1 core-hours for P_{19}, 236.0 for P_{23} and 1,292.7 for $\Delta_6 - 5$ cover solving *and* checking. Portable root values, which is exactly the check Section 5.2 supports, can be easily compared by anyone replicating our computation.

For P_{19} at unit margin the instance is the dissent-Boolean encoding of Appendix A.1 at $q = 19$, $k = 5$: 855 variables and 12,255 clauses. The cubes are the 142,251 base states of the 6-vertex base $\{1, 2, 3, 4, 5, 12\}$, of which 22,876 survive the arc-orbit break, and each is handed whole to CaDiCaL with `--lrat=true --checkproof=0`. Each LRAT proof is checked by `lrat-trim`. The runtime was 2.64 hours on ten cores. Its coverage half is the same construction as the one below, at 143,032 clauses, and solves in 7.56 seconds.

For the P_{23} refutation of Section 3.1, the base is $\{1, 2, 3, 4, 6, 7\}$ with 3,414,729 base states, of which 343,896 survive the arc-orbit and non-arc-orbit breaks; every one is unsatisfiable, with an LRAT proof independently verified by `lrat-trim`. The runtime was 21.72 hours on eleven cores. The coverage half of the proof is a single unsatisfiable instance of 350 constraint clauses, 356 symmetry-breaking clauses and 3,414,729 negated cubes, one per base state, since coverage must see *every* base state whereas the orbit breaks apply to the search half only. It solves in just under 20 minutes on one core and emits a 2.05 GiB LRAT proof.

For the $\Delta_6 - 5$ refutation of Section 3.2 the instance is the dissent-Boolean encoding at $n = 22$ and plain majority: 1,155 variables and 17,710 clauses. The tournament has no non-trivial automorphism, so there is no orbit break, and all 742,374 base states of the six-vertex base $\{1, 2, 3, 4, 13, 19\}$ (in the labelling of Δ_6) are cubes; every one is unsatisfiable, with an LRAT proof independently verified by `lrat-trim`. The runtime was about 118 hours on eleven cores. The coverage half is the same construction as for P_{23}, with 742,374 negated cubes, and solves in 82.8 seconds on one core. This refutation is the most expensive of the three, although its tournament is the smallest: with no automorphisms there is nothing to break, so every base state reaches the solver, against one in ten for P_{23}.

B.2 What the certificates do and do not establish

The trust chain has four links:

link	status
every live cube is unsatisfiable	machine-checked: CaDiCaL, rechecked by `lrat-trim`
the cubes cover every lex-canonical base assignment	machine-checked: LRAT, verified by `lrat-trim`
(L1) orbit anchoring loses no generality	human-checked: Lemma 2.1
(L2) lex-ordering the voters loses no generality	human-checked: Lemma 2.3

The two human-checked links are proved in Section 2. Lemma 2.1 is what licenses solving only the *live* cubes, in its arc/non-arc form: $\mathrm{Aut}(P_q)$ is transitive on arcs and on non-arcs, so a voter's top pair may be assumed

to be a fixed arc or a fixed non-arc. Lemma 2.3 licenses the lex chain in the coverage computation. For $\Delta_6 - 5$ the chain is one link shorter: with no automorphism to exploit, no orbit break is applied and Lemma 2.1 is not used, so only Lemma 2.3 remains human-checked. The lemmas are human-checked links not because they are doubtful but because nothing in the pipeline checks them; the depth-first engine inherits them, but shares no implementation with this one, so their agreement is evidence about the implementations, not the lemmas.

Appendix C. The order-12 case analysis

Every tournament on 12 vertices is 5-inducible, so $N(5) \geq 13$. This appendix describes that computation. It is one of the results in this paper that was not run on the laptop of Section 3: it ran on a shared cluster, and its runtime is reported in cluster core-hours (see Appendix A.5).

C.1 Reduction to one-vertex extensions

A direct census is out of reach: $D_{12} \approx 1.5 \times 10^{11}$ classes, 170 times the order-11 census of Chindelevitch and Harutyunyan (2026). We use that census as the outer loop instead. Let us write $L + S$ for the tournament obtained from an order-11 tournament L by adding a vertex v whose out-neighbourhood is $S \subseteq V(L)$. Two observations simplify the analysis.

(C1) Every order-12 tournament is $L + S$ for some order-11 L and some S.

(C2) Restricting a profile that realises $L + S$ to $V(L)$ gives a profile that realises L.

By (C1), enumerating all D_{11} order-11 classes and settling all $2^{11} = 2{,}048$ extensions of each settles every order-12 tournament; all of the former are 5-inducible by the census of Chindelevitch and Harutyunyan (2026), which is what makes the outer loop available. By (C2), a profile realising $L + S$ may be sought by *inserting* v into a profile realising L with no loss of generality, since every profile for $L + S$ arises that way. That is what phase 1 exploits.

C.2 Three phases

The census is organised into three phases, with the fastest ones occurring earlier.

Phase 1, the covering search. Rather than 2,048 separate instances per order-11 tournament, one search enumerates profiles realising L *at unit margin* and, for each, records every out-neighbourhood obtainable by inserting v into its five orders: v's position in each order is a threshold, and the support of $v \to u$ is the number of orders that place v above u, so one traversal certifies many extensions at once. This is the engine of Appendix A.2 with v's placement read off rather than fixed in advance. The margin makes this phase pay twice: it keeps the enumeration of profiles for L small, and a unit-margin witness is in particular a majority witness, so every extension the phase covers is settled in both regimes at once. The phase is one-sided: an extension it covers is certified 5-inducible, but one left over is not refuted. This phase settles every one of the 2,048 extensions of 99.5% of the order-11 tournaments.

Phase 2, deciding what is left. Each extension phase 1 leaves open is materialised as an order-12 tournament $L + S$ and decided by the same engine, again at unit margin, where a witness settles it positively. About one extension in 200,000 reached this phase, and every one of them is 5-inducible with unit margin.

Phase 3, plain majority. Only a *majority* refutation would be a counterexample, so any instance negative at unit margin is re-run without the margin restriction. Phase 3 was never invoked: every tournament was settled at unit margin, so the weaker question never had to be asked.

C.3 One tournament of each converse pair

If $\pi_1, \ldots, \pi_5$ realise T, their reversals realise the converse $\bar{T}$, with the same margins, so T is 5-inducible if and only if $\bar{T}$ is. Moreover $\overline{L+S} = \bar{L} + (V(L) \setminus S)$, so settling all 2,048 extensions of L settles all 2,048 of $\bar{L}$ as well. We therefore keep one tournament of each converse pair, by the rule that L is kept if and only if $\mathrm{canon}(L) \leq \mathrm{canon}(\bar{L})$. This keeps exactly one of each pair together with every self-converse tournament,

$(D_{11} + S_{11})/2$ in all, where S_{11} is the number of self-converse order-11 classes; this saves a factor of $\lessapprox 2$. We also check the number of tournaments kept is $(D_{11} + S_{11})/2$.

Total runtime: approximately 10,000 cluster core-hours. The search code, the per-part records and the roll-up are at `cluster/jz_n12cover` in the repository of Section 5.

Appendix D. Exact enumeration counts

In this table, D_n, R_n and S_n are the numbers of isomorphism classes of all, of regular, and of self-converse tournaments on n vertices. They match OEIS A000568 (OEIS Foundation Inc. 2026a), A096368 (OEIS Foundation Inc. 2026c), and A002785 (OEIS Foundation Inc. 2026b).

n	D_n	R_n	S_n
11	903,753,248	1,223	279,968
12	154,108,311,168	—	1,492,288
13	—	1,495,297	95,458,560
15	—	18,400,989,629	—

Appendix E. Locally transitive tournaments, at unit margin, for every odd $k \geq 3$

A tournament is *locally transitive* if for every vertex v both $N^+(v)$ and $N^-(v)$ induce transitive subtournaments. Equivalently it is *round*: the vertices admit a cyclic order $v_0, \ldots, v_{n-1}$ in which every $N^-(v) \cup \{v\}$ and $\{v\} \cup N^+(v)$ is an interval, so that, orienting the cyclic order along the arcs, each vertex beats exactly the $d_i = |N^+(v_i)|$ vertices that follow it: $N^+(v_i) = \{v_{i+1}, \ldots, v_{i+d_i}\}$, indices modulo n. The equivalence is due to Brouwer (1980), who proved it in the course of enumerating the class, and to Alspach and Tabib (1982), who state it for tournaments under the name *domination orientable*; Bang-Jensen later generalised it to digraphs.

Theorem E.1. *Every locally transitive tournament is 3-inducible with unit margin, and hence k-inducible with unit margin for every odd $k \geq 3$.*

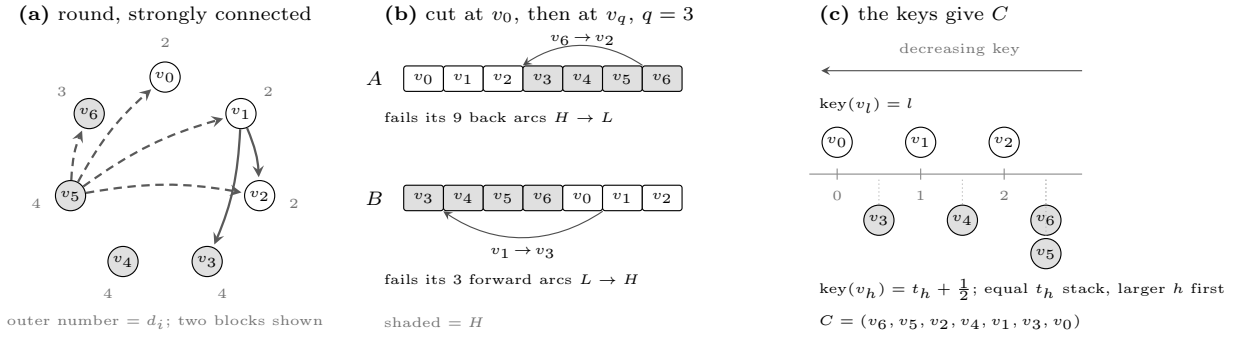


Figure E.1: The construction of Appendix E on a round tournament with out-degree sequence $(2, 2, 2, 4, 4, 4, 3)$, for which $q = 3$. **(a)** The cyclic order, each v_i labelled with its out-degree d_i, so that its out-neighbours are $v_{i+1}, \ldots, v_{i+d_i}$; the two blocks drawn are those of $v_1 \in L$ and $v_5 \in H$, and that of v_5 wraps, with back-targets v_0, v_1, v_2. **(b)** A and B are the same cyclic order cut at v_0 and at v_q, each failing exactly the arcs that cross its own cut, 9 back arcs and 3 forward arcs; one of each is drawn. **(c)** The keys, C being the vertices in decreasing order of key. A vertex of L sits at its index and one of H half a unit past its ceiling, so reading leftwards puts v_h before v_l exactly when $v_h \to v_l$; v_5 and v_6 share a ceiling and the tie-break separates them, giving $C = (v_6, v_5, v_2, v_4, v_1, v_3, v_0)$.

A locally transitive tournament that is not strongly connected is transitive: a strongly connected component C with $|C| \geq 3$ contains a directed triangle, and any vertex outside C either beats all of C or loses to all of it, contradicting local transitivity. A transitive tournament is 1-inducible at margin 1, its single voter supporting

every arc. Adding an order together with its reverse adds one supporter and one opponent to each arc, raising every support by 1 and k by 2 and so leaving every margin $2c(e) - k$ unchanged. Padding with the identity order $(1, 2, \ldots, n)$ and its reverse therefore gives unit margin at $k = 3$, and repeating gives every odd k.

Let T be strongly connected and round, and let d_i be the out-degree of vertex i; Figure E.1 illustrates what follows, on seven vertices. Cutting the cyclic order at v_0 gives the linear order $A = (v_0, \ldots, v_{n-1})$. We write $r_i = i + d_i$, an integer deliberately *not* reduced modulo n. For each v_i, the out-neighbours are then $v_{i+1}, \ldots, v_{r_i}$ with indices reduced only when they are read as vertices. The sequence r is non-decreasing, since if $r_{i+1} < r_i$ then $v_{r_{i+1}+1}$ is an out-neighbour of v_i, while its own interval includes v_i, a contradiction. Let $H = \{i : r_i \geq n\}$ be the vertices with an out-neighbour that the cut defined by A placed *before* them. As r is non-decreasing, $H = \{q, \ldots, n-1\}$ for some q; write $L = \{0, \ldots, q-1\}$. Neither set is empty: $H = \emptyset$ would make v_{n-1} a sink and $L = \emptyset$ would force $d_0 \geq n$, so neither occurs. For $h \in H$ put $t_h = r_h - n$, so that the out-neighbours of v_h placed before it by the cut are exactly $v_0, \ldots, v_{t_h}$: we call them the *back-targets* of v_h, and t_h its *ceiling*. By monotonicity of r, $t_q \leq \cdots \leq t_{n-1}$. Every back-target lies in L, that is $t_{n-1} < q$: otherwise $v_{n-1} \to v_q$, while $r_q \geq n$ implies $v_q \to v_{n-1}$. The arcs between L and H therefore divide into the *back arcs* going from H to L, and the *forward arcs* going from L to H; and for $h \in H$ and $l \in L$ we have $v_h \to v_l$ exactly when $l \leq t_h$.

We give each vertex a *key*: $\mathrm{key}(v_l) = l$ for $l \in L$ and $\mathrm{key}(v_h) = t_h + \frac{1}{2}$ for $h \in H$, ties among equal ceilings broken in favour of the larger index, and let C list the vertices in decreasing order of key. Integer and half-integer keys never collide, so that tie-break is the only one needed, the keys are pairwise distinct and C is a linear order; and for $h \in H$ and $l \in L$, $\mathrm{key}(v_h) > \mathrm{key}(v_l) \iff t_h + \frac{1}{2} > l \iff l \leq t_h \iff v_h \to v_l$, so C places v_h before v_l exactly when $v_h \to v_l$. We now take the three orders $A = (v_0, \ldots, v_{n-1})$, $B = (v_q, \ldots, v_{n-1}, v_0, \ldots, v_{q-1})$, which is the same cyclic order cut at v_q instead of at v_0, and C. Every arc is now easily seen to have support exactly two. *Arcs with both endpoints inside L or inside H* are only supported by A and B; *forward arcs* $v_l \to v_h$ are only supported by A and C; and *back arcs* $v_h \to v_l$ are only supported by B and C, so the profile realises T at unit margin. The padding argument above extends this margin to any odd $k \geq 3$. ■

Appendix F. Criticality at 21 vertices

This appendix carries the detail behind Section 3.6. Throughout, h ranges over the four unit-margin obstructions among the 110 vertex-transitive tournaments on 21 vertices (Section 3.7).

Vertex-critical but maximally not arc-critical. Arc-criticality implies vertex-criticality, and the converse fails here as widely as it can. Each h is *vertex-critical* at unit margin, and for h_1 every single-arc reversal remains a unit-margin obstruction.

Arc-criticality is a property of the arc, not just the tournament. All four have ten arc orbits of size 21; we sweep one representative per orbit at unit margin and get results that differ by tournament *and* by arc, `U` marking a reversal that is still not unit-margin inducible and `S` one that is. h_2 splits five orbits to five, so exactly half its arcs are critical and we call it *arc-semi-critical*; with P_{23} arc-critical unrestricted and h_1 critical nowhere, all three possibilities occur among our tournaments. Intriguingly, every one of the 15 reversals that remains an obstruction belongs to a tournament with *cyclic automorphism group*, while all 20 orbits of the two with the nonabelian group of order 21 are inducible.

tournament	Aut	the ten orbits	obstructions
h_1	$\mathbb{Z}_{21}$	`U U U U U U U U U U`	10
h_2	$\mathbb{Z}_{21}$	`U S U S S U U S S U`	5
h_4, h_5	$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	`S S S S S S S S S S`	0

Appendix G. The search below 22 vertices

This appendix summarises the search for an obstruction on fewer than 22 vertices, which would improve the upper bound of Section 3.2. It has found none.

The search runs downward, because 5-inducibility is inherited by subtournaments: restricting a witness for T to a subset of the vertices gives a witness for the subtournament. An obstruction on fewer vertices therefore lies below some larger one, and the natural place to look is among the vertex deletions of known obstructions. A host is settled positively as soon as one base state yields a witness, which is cheap, and negatively only by a complete search, which is not; so each level is first screened for witnesses, and only hosts that resist the screen are searched further.

The starting points are the 36 doubly regular tournaments on 23 vertices other than P_{23}, all of which are obstructions (Section 3.7). P_{23} itself is excluded, since $P_{23} - v$ is 5-inducible (Section 3.6) and so, by inheritance, is every tournament below it. Up to isomorphism and up to taking the converse, the levels below the 36 are as follows.

vertices	deleted	classes	status
22	1	378	2 searched completely, both obstructions (Section 3.2)
21	2	4,108	1,173 settled, all 5-inducible
20	3	28,724	all 5-inducible

At 22 vertices all 378 classes were first screened at up to 50 base states each, without a single witness, and the two complete searches were chosen from among them. At 21 vertices the 1,173 settled classes include all 506 two-vertex deletions of Δ_6 and Δ_{28}, so every 21-vertex subtournament of either parent is 5-inducible, and in particular both 22-vertex obstructions are vertex-critical. The other settled classes are a stratified random sample of 400, all 53 classes below Δ_{11}, and every class obtained by deleting vertex 5 together with one other vertex, the deletion that produced both obstructions; the remaining 2,935 classes are open. At 20 vertices every class is 5-inducible, and so, by inheritance, is every tournament on 19 vertices or fewer below the 36.

Witnesses become rarer as one climbs. At 21 vertices about one base state in 110 yields a witness; at 22 vertices the screen exhausted 13,863 base states and found none. A host at 21 vertices therefore costs a few dozen cluster core-hours to settle, and an obstruction at 22 about 1,500 to prove.

Two obstructions on 25 vertices from other families were descended the same way: P_{27} with two vertices deleted, and the circulant tournament on 25 vertices in which every arc lies in six directed triangles. Neither is 5-inducible, by complete searches, and every 21-vertex subtournament of either is 5-inducible, so neither family yields an obstruction below 22.